

- Diseases of the upper respiratory tract (nose, sinuses, nasopharynx, & larynx)
- Diseases of the lower respiratory tract
 - Brief review of the normal lung
 - Congenital anomalies
 - Atelectasis
 - Chronic obstructive pulmonary diseases
- Pulmonary infections
- Diffuse interstitial (infiltrative, restrictive) diseases
- Tumors

Pathophysiology of common diseases of upper respiratory tract

- Include the nose, pharynx, and larynx and their related parts.
- Inflammatory diseases, mostly in the form of the common cold, are the most common disorders of the nose and accessory air sinuses.
- Most of these inflammatory conditions are viral in origin, but can be complicated by superimposed bacterial infections

Inflammations

Infectious Rhinitis.

- “common cold,”
- caused by one or more viruses.
- Major offenders are adenoviruses, echoviruses, and rhinoviruses.
- During the initial acute stages, the nasal mucosa is thickened, edematous, and red; the nasal cavities are narrowed; and the turbinates are enlarged.
- These changes may extend, to produce pharyngotonsillitis.
- Secondary bacterial infection enhances the inflammatory reaction and produces a mucopurulent or sometimes frankly suppurative exudate.

Allergic Rhinitis.

- It is an IGE mediated immune reaction i.e initiated by sensitivity reactions to allergens
- The allergic reaction is characterized by marked mucosal edema, redness, and mucus secretion, accompanied by a leukocytic infiltration in which eosinophils are prominent.
- is usually complicated by ***Nasal polyps*** , it is focal protrusion of mucosa which can ulcerate, infected or can impair sinus discharge

Nasal Polyps

- Recurrent attacks of rhinitis
- On histologic examination these polyps consist of edematous mucosa having a loose stroma, often harboring hyperplastic or cystic mucous glands, infiltrated with a variety of inflammatory cells, including neutrophils, eosinophils, and plasma cells with occasional clusters of lymphocytes.
- When multiple or large, the polyps may encroach on the airway and impair sinus drainage.
- only 0.5% of atopic patients develop polyps

Chronic Rhinitis

- Chronic rhinitis is a sequel to repeated attacks of acute rhinitis, either microbial or allergic in origin, with the eventual development of superimposed bacterial infection.
- there is superficial desquamation or ulceration of the mucosal epithelium and a variable inflammatory infiltrate
- These suppurative infections sometimes extend into the air sinuses.

Sinusitis

- Commonly preceded by rhinitis
- Impairment of drainage of the sinus by inflammatory edema of mucosa
- Etiologies are the normal flora of oral cavity
- Complications include osteomyelitis & septic thrombophlebitis
- severe forms of chronic sinusitis are caused by fungi (e.g., mucormycosis), especially in patients with diabetes.

Pharyngitis and tonsillitis (tonsilopharyngitis)

- Frequent concomitants of viral infections
- Commonest bacterial offender is beta hemolytic streptococcus sometimes s.aureus
- Pseudomembrane – exudative
- May have a late sequale
- A typical appearance is of enlarged, reddened tonsils (due to reactive lymphoid hyperplasia) dotted by pinpoints of exudate emanating from the tonsillar crypts, socalled *follicular tonsillitis*

Tumors

- Tumors in these locations are infrequent, arise from the mesenchym or epithelium
 - Nasopharyngeal angiofibroma
 - Sinonasal papillomas
 - Olfactory neuroblastomas
- **Nasopharyngeal carcinoma** – may take one of the three patterns
 - Keratinizing squamous cell carcinoma
 - Non-Keratinizing squamous cell carcinoma
 - Undifferentiated carcinoma

Nasopharyngeal ca

✖ Three factors affect its origin

- Heredity
- Age
- Infection with EBV

- Common in Africa
- Grow silently until they spread to LNs
- Unlike keratinized, the undifferentiated ca is the most radiosensitive

Larynx

- The most common disorders that affect the larynx are inflammations, tumors uncommon

Inflammations

- ***Laryngoepiglottitis***
- caused by respiratory syncytial virus, Haemophilus influenzae, or β -hemolytic streptococci may induce such sudden swelling of the epiglottis and vocal cords as to constitute a medical emergency.
- This form of disease is uncommon in adults because of the larger size of the larynx and the stronger accessory muscles of respiration.

Ctd...

Croup

- Laryngeotracheobronchitis in children
- Narrowing of airways produces inspiratory stridor, **barking cough**
- ***Laryngitis in smokers***
- Predisposition to squamous epithelial changes

Ctd...

Vocal cord nodules & polyps

- In heavy smokers & Singers
- Singers nodules are bilateral whereas polyps are unilateral
- Because of their strategic location and accompanying inflammation, they characteristically change the character of the voice and often cause progressive hoarseness.
- They virtually never give rise to cancers.

Carcinoma of larynx

- Hyperplasia, dysplasia, carcinoma in situ, & invasive carcinoma
- SCC – 95%, rarely adenocarcinoma
- Manifested by persistent hoarseness, dysphagia, and dysphonia
- most commonly seen in men, related to smoking and alcohol use, in the sixth decade of life

Acute otitis media

- Acute inflammation in middle ear
- < 3 weeks (month)
- Often associated with a viral upper respiratory infection
- Most common reason for medical therapy for children younger than 5 years
- Most children have at least one episode of AOM (by age 3, 50-85%)
- Peak incidence age 6-15
- Only 20% are adults

Trigger:

- Allergies
- Upper respiratory tract infections
- GER (especially children)
- Adenoid hypertrophy
- Viral (30-70%) (RSV,Rhinovirus ,Coronavirus. Influenza, parainfluenza)
- Bacterial (55%) Streptococcus pneumoniae (44%), Haemophilus influenzae (41%) • Combined (15%)
- Otalgia (not always), Fever. Hearing loss (speech delay for children)

THE END